

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Anti-seize compound, Part Number 3824879, or equivalent.

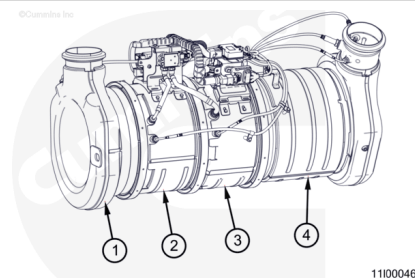
Additional Service Items

- Marking pen
- Jack stands, or equivalent
- Scissor jack, or equivalent.

General Information

⚠ WARNING ⚠

The material captured in a partial flow diesel particulate filter and/or a diesel particulate filter may contain elevated concentrations of metals. Primarily zinc, molybdenum, and possibly polynuclear aromatic hydrocarbons, that may be regulated. These materials must be characterized, handled, and disposed of according to applicable local regulations. In addition, due to the presence of the above-listed chemicals and other potentially toxic components such as oxides of calcium, zinc, phosphorous, silicon, sulfur, and iron, exhaust filter maintenance must be completed only by appropriately trained personnel.



⚠ CAUTION ⚠

The aftertreatment diesel particulate filter element is made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst elements can result.

Due to the number of exhaust aftertreatment applications, this procedure has been written to be generic. **Not** all illustrations within this document will represent the application being serviced.

The aftertreatment exhaust assembly is composed of four sections. These sections are:

1. The aftertreatment diesel oxidation catalyst (DOC)
2. The aftertreatment diesel particulate filter (DPF)
3. The decomposition reactor tube
4. The selective catalytic reduction (SCR).

Note : If the aftertreatment DPF is being replaced due to progressive damage from engine oil or fuel, clean the exhaust pipe from the turbocharger outlet to the aftertreatment DOC.

Note : If a malfunction resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 (</qs3/pubsys2/xml/en/bulletin/4021600.html>).

Note : If a malfunction resulted in coolant entering the exhaust system, perform the Aftertreatment Diesel Particulate Filter Restriction Test to determine if the DPF needs to be replaced. Refer to Procedure 011-085 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-085.html>)

Note : Do not steam clean the inlet of the DOC.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

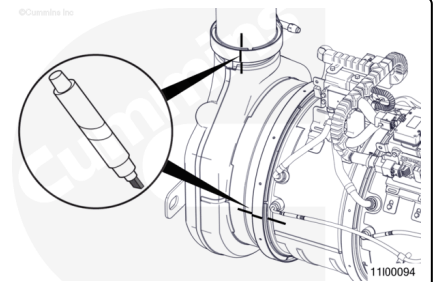
During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

- Disconnect the battery cables. See equipment manufacturer service information.
- Disconnect the original equipment manufacturer (OEM) wiring harness from the aftertreatment interface harness. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Remove the exhaust gas temperature sensor probes from the aftertreatment DOC and DPF. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-449.html>)
- Remove the inlet differential pressure sensor probe from the DPF. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-047.html>)

Remove

⚠ WARNING ⚠

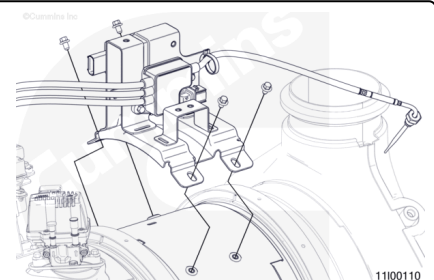
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Draw an orientation reference line across each of the V-band clamps, aftertreatment canister sections, sensor mounting bracket strap, and connection points to the exhaust pipe. This aids in lining up the sections and the V-band clamps to their original orientation during installation.

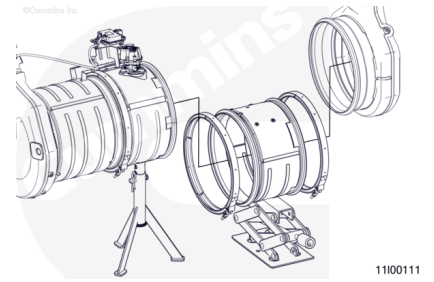
Remove the capscrews that secure the sensor mounting bracket to DPF. Refer to Procedure 011-141 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-141.html>)

Support the aftertreatment DPF sensor mounting bracket after detaching it from the DPF.



⚠ CAUTION ⚠

Do not use an air tool to remove the V-band clamp nut. An air tool will damage the threads. Apply thread lubricant to the V-band clamp threads prior to nut removal.



Note : Depending upon aftertreatment system installation, the full aftertreatment system may need to be removed from the vehicle prior to disassembly. Refer to Procedure 011-050 in section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-050.html>)

Support the aftertreatment decomposition tube to prevent it from being freely suspended after the aftertreatment DPF is removed.

Support the aftertreatment DPF with a lowering jack.

Remove the V-band clamps from the inlet and outlet flanges of the aftertreatment DPF.

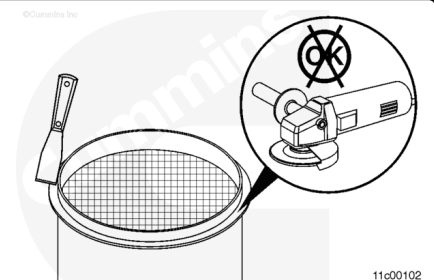
Discard the gaskets from the V-band joint.

Note : It may be necessary to remove additional mounting hardware to remove the aftertreatment DPF from the vehicle.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use a grinder to remove residual gasket material as this may damage the flange and cause the connection to leak.



⚠ CAUTION ⚠

Do not use an open flame to burn off soot accumulation from the face of the aftertreatment diesel particulate filter.

⚠ CAUTION ⚠

Do not scrape off soot accumulation from the face of the aftertreatment diesel particulate filter.

Note : It is **not** possible to see light straight through the holes in the aftertreatment DPF.

Note : If the aftertreatment DPF is being replaced due to progressive damage from engine oil or fuel, clean the exhaust pipe from the turbocharger outlet to the aftertreatment DOC. Refer to Procedure 011-049 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-049.html)

Remove any residual gasket material from the flanges on the aftertreatment DPF with a scraping tool.

Note : Avoid dropping fragments of gasket material into the aftertreatment DPF.

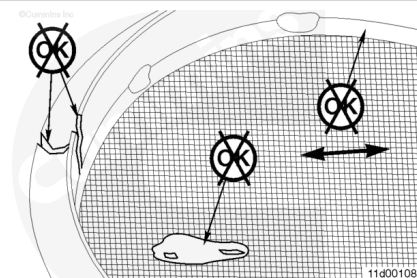
If the aftertreatment DPF has been removed for any reason, refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600.

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Replace the aftertreatment DPF if damage is found.

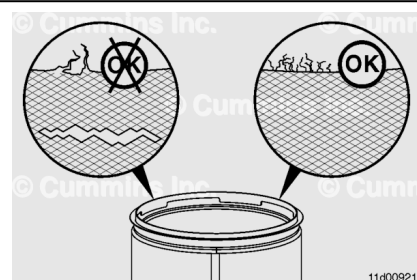
If the aftertreatment DPF has shifted, moved, or is loose inside of the canister, replace the aftertreatment DPF.

If the aftertreatment DPF canister is cracked or otherwise damaged, replace the aftertreatment DPF.



Inspect the DPF body joint insulation for visible wear at the DPF inlet and outlet.

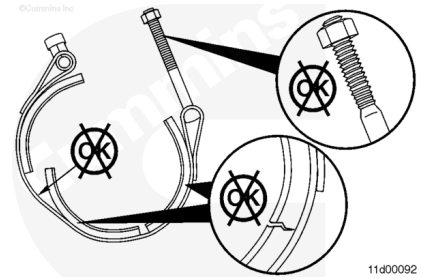
Replace the DPF if excessive fraying or any cuts are observed.



Inspect the V-band clamps and mounting straps for signs of over-extension. The band **must not** be bent or damaged.

Inspect the V-band clamps and mounting strap threads for damage.

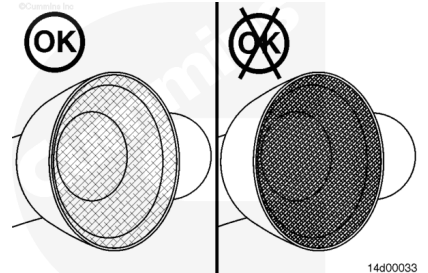
Replace the V-band clamp or strap if damage is found.



Note : Some accumulation of exhaust residue/soot is normal. It does not indicate a malfunctioning aftertreatment DPF.

A heavy buildup of exhaust residue/soot can indicate a malfunctioning aftertreatment DPF.

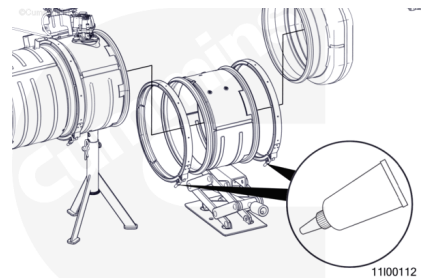
If the aftertreatment DPF is being replaced due to black smoke and the exhaust stack is found to have heavy black soot accumulation, clean the last 152 to 254 mm [6 to 10 in] of the exhaust system outlet.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use an air tool to tighten the V-band clamp. An air tool will damage the threads.

Service locations in California. See Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787 (</qs3/pubsys2/xml/en/bulletin/6642787.html>).

Install new gaskets on the inlet and outlet of the aftertreatment DPF canister.

Apply a coat of anti-seize compound, Part Number 3824879, or equivalent, to the threads of the V-band clamps.

Install the aftertreatment DPF.

Tighten the V-band clamps.

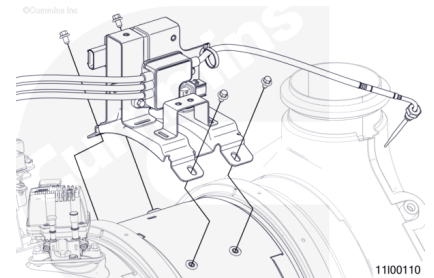
Torque Value: 20 n•m [177 in-lb]

Note : Do not place the T-bolt section of the V-band clamp near the wiring harness.

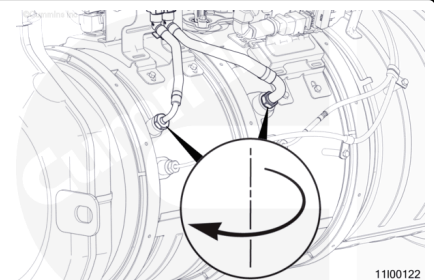
Install the insulation between the DPF and the sensor support bracket.

Install the sensor mounting bracket. Refer to Procedure 011-141 in Section 11.

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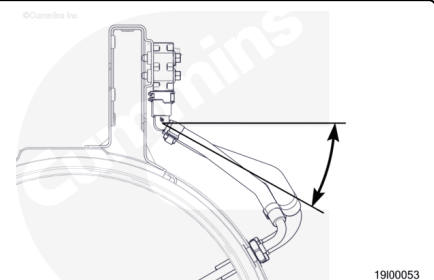


Install the aftertreatment DPF pressure sensor tubes into the aftertreatment DPF. Refer to Procedure 011-047 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)

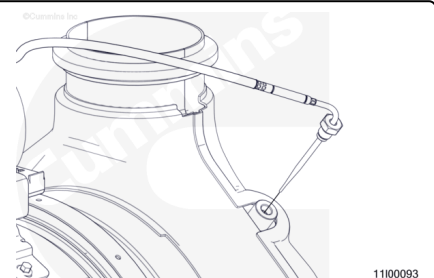


The DPF differential pressure sensor tubes **must** have a continuous downward slope away from the sensor ports whether in a vertical or horizontal aftertreatment configuration. A 15 degree sensor tube down angle is recommended to make sure no cavity in the tubes collect moisture. Refer to Procedure 011-047 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)

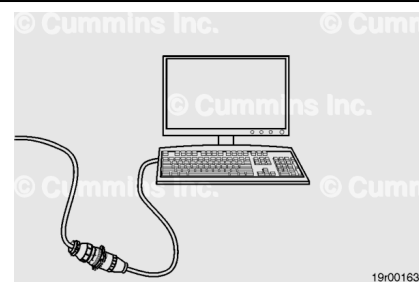


Install the exhaust gas temperature sensor probes into the aftertreatment DOC and DPF. Refer to Procedure 019-449 in Section 19. (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)



Adjust

If the aftertreatment DPF was replaced for high soot loading or plugging, it will be necessary to reset the stored soot load in the engine control module (ECM). This can be performed by using INSITE™ electronic service tool to perform the Aftertreatment Maintenance Reset procedure under the Advanced ECM Data menu.



An aftertreatment DPF that is replaced will require use of the Aftertreatment Filter Installation function. This function is found in INSITE™ electronic service tool Advanced ECM Data menu.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the battery cables. See equipment manufacturer service information.
- If the aftertreatment DPF was replaced, perform a stationary regeneration to verify and finalize the repair. If any fault codes become active during a stationary regeneration, reference the appropriate Fault Code Troubleshooting Manual.
- Operate the engine and check for exhaust leaks and proper operation.